North Carolina State University Invitational - Repaired practice edition

Division B | Botany | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

4. (1 points) Authored replacement for source item 4: A supplied diagnostic report identifies Colletotrichum, a fungal genus, as the cause of a plant infection named anthracnose. Is this diagnosed infection fungal or bacterial? Explain using the report.

Answer: _____
_____
_____

9. (1 points) Authored replacement for source item 9: Supplied crop survey: soybeans 40 hectares, corn 30, cotton 20. Which of these genetically modified crop plantings occupies the largest area in this survey?

Answer: _____
_____
_____

14. (1 points) Authored replacement for source item 14: A plant germinates, flowers, sets seed and dies within one growing season. Classify its life cycle as annual, biennial or perennial.

Answer: _____
_____
_____

21. (1 points) Authored replacement for source item 21: Compare supplied ages of individual trees: A=600 years, B=1,200 years, C=4,000 years. Which individual is oldest? Explain why this comparison does not establish the age of a clonal colony.

Answer: _____
_____
_____

57. (1 points) Authored replacement for source item 57: The repair defines its reservoirs as sedimentary rocks, the atmosphere, living forest biomass and ice. Supplied carbon stores are 80, 10, 8 and 2 arbitrary units respectively. Which reservoir stores the most carbon in this model?

Answer: _____
_____
_____

60. (1 points) Authored replacement for source item 60: An animal can digest starch but cannot hydrolyze the beta-1,4 glycosidic links in cellulose because it lacks cellulase. Explain why “it cannot break any glycosidic bonds” is too broad.

Answer: _____
_____
_____

66. (1 points) Authored replacement for source item 66: A crop receives an inserted gene using a laboratory transformation method. Another crop is selected only through conventional crossing. Which is genetically engineered by the stated method?

Answer: _____
_____
_____